

VP150 Recitation Class 7

Yangrui Meng

July 18th, 2024

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 - Definition
 - Rolling without slipping
- 2 Work and Conservation of Angular Momentum
 - Work in Rotational Motion
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- 3 The Gyroscopic Effect
- 4 Equilibrium

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Definition

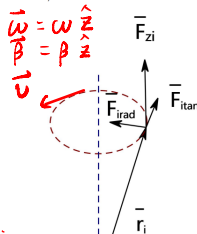
Definition

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Rotation with Fixed Axis

The net force can be decomposed into three components: radial, tangential and along the z-axis $\vec{F}_i = \vec{F}_{i,rad} + \vec{F}_{i,tan} + \vec{F}_{i,z}$

$$\vec{F}_i = \vec{F}_i = \underbrace{\vec{F}_{i,rad}}_{\text{radial}} + \underbrace{\vec{F}_{i,tan}}_{\text{tangential}} + \underbrace{\vec{F}_{i,z}}_{\text{along the z-axis}}$$



the only component that contributes to rotation
if the motion is in a plane $\Rightarrow \omega, \beta$ along z axis

Dynamics for Rotation

Formula

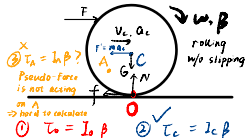
Formula:

$$\tau_z = I_z \varepsilon_z \quad \Rightarrow \quad f = ma$$

Translation	Rotation
m	I
a	ε
v	ω
F	τ

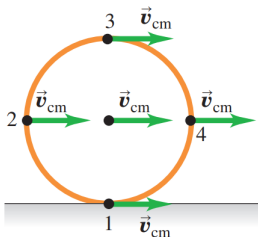
Rolling without slipping

Often use the CoM as rotational axis

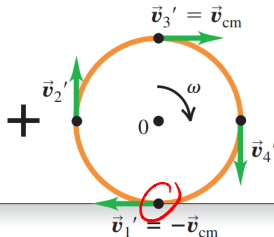


Relationship between velocity of CoM and angular velocity

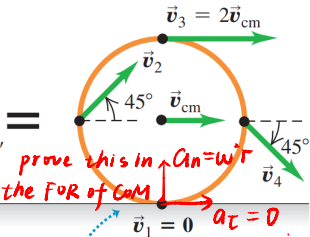
Translation of center of mass:
velocity $\vec{v}_{cm}$



Rotation around center of mass:
for rolling without slipping,
speed at rim = v_{cm}



Combined motion



Pseudo-force is seen as acting on CoM.

v & ω ?

use the condition:
contact point has zero
relative velocity

$$v_{cm} - \omega r = 0$$

$$\Rightarrow v_{cm} = \omega r$$

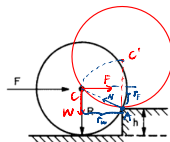
$$\frac{d}{dt} \Rightarrow a_{cm} = \alpha r$$

Wheel is instantaneously at rest
where it contacts the ground.

CANT use $f = \mu N$
for pure rolling

Exercise

Question 4. (5 points) What is the magnitude of a horizontal force $\mathbf{F}$ (applied at the axle) able to push a wheel of weight w and radius R over a perfectly rigid step of height h ?



Choose A as axis:

$$r_F = R - h$$

$$r_w = \sqrt{R^2 - (R - h)^2}$$

$$F \cdot (R - h) \geq w \cdot \sqrt{R^2 - (R - h)^2}$$

$$\Rightarrow \underline{F \geq \frac{\sqrt{2Rh - h^2}}{R - h} w}$$

Does this make sure it climb up the step eventually?

$$r_F \uparrow \quad r_w \downarrow \quad \Rightarrow \quad F \cdot r_F \geq w \cdot r_w$$

Exercise

A bowling ball rolls without slipping down a ramp that is inclined at an angle β to the horizontal (**Fig. 10.19a**). What are the ball's acceleration and the magnitude of the friction force on the ball? Treat the ball as a uniform solid sphere, ignoring the finger holes.

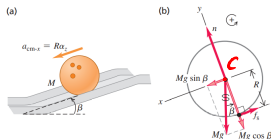
$$I = ?$$

$$\begin{cases} ma = mg \sin \beta - f_s & \leftarrow \text{force} \\ f_s R = I_c \alpha & \leftarrow \text{torque} \\ a = \alpha R & \leftarrow a-v \text{ relation} \end{cases}$$

$$\Rightarrow a = \frac{5}{7} g \sin \beta$$

$$f = \frac{2}{7} mg \sin \beta$$

10.19 A bowling ball rolling down a ramp.



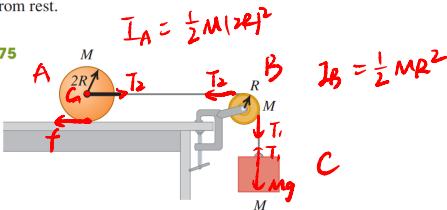
Handwritten notes and diagrams for the exercise:

- A diagram of a circle representing the ball, with center C and point A at the bottom. The velocity of the center is v_c and the angular velocity is ω . The acceleration of the center is a_c .
- The condition for rolling without slipping is written as $v_A = v_c - \omega R = 0$.
- The equation $v_c = \omega \cdot R$ is circled.
- The derivative of the velocity condition is $\Rightarrow \frac{dv_c}{dt} = \frac{d\omega}{dt} \cdot R$.
- The final equation is $\Rightarrow a_c = \alpha R$.

Exercise

10.75 • A uniform solid cylinder with mass M and radius $2R$ rests on a horizontal tabletop. A string is attached by a yoke to a frictionless axle through the center of the cylinder so that the cylinder can rotate about the axle. The string runs over a disk-shaped pulley with mass M and radius R that is mounted on a frictionless axle through its center. A block of mass M is suspended from the free end of the string (Fig. P10.75). The string doesn't slip over the pulley surface, and the cylinder rolls without slipping on the tabletop. Find the magnitude of the acceleration of the block after the system is released from rest.

Figure P10.75



$$A: \begin{cases} f \cdot 2R = I_A \epsilon_A \\ T_2 - f = M a_A = M a \end{cases}$$

$$B: (T_1 - T_2) \cdot R = I_B \epsilon_B$$

$$C: Mg - T_1 = M a_C = M a$$

$$a = \epsilon_A \cdot 2R$$

$$a = \epsilon_B \cdot R$$

unknown:

$T_1, T_2, f, a, \epsilon_A, \epsilon_B$

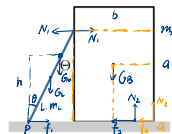
6 eq. 6 unknown

$\Rightarrow \dots$

Exercise

Problem 3. (15 points) A man of mass m_M climbs up an unsecured ladder of length L and mass m_L . It is resting on the side of a large empty box of height a and width b , which has mass m_B . There is no friction between the ladder and the box.

- If the ladder makes an angle of θ degrees to the vertical side of the box, find the coefficient of friction between the box and the ground needed to prevent the box from sliding away from the ladder when the man stands at a height h above the ground.
- Assuming the friction is sufficient to prevent the box from sliding, find the maximum height the man can climb before the box begins to tip over.



The acceleration due to gravity g is known.

(a)

For B:

$$\text{Force: } \begin{cases} N_2 = G_B \\ N_1 = f_1 \\ f_2 \leq \mu_s N_2 \end{cases}$$

For L:

$$\text{Torque: } N_1 L \cos \theta = G_L \cdot \frac{L}{2} \sin \theta + G_M \cdot h \tan \theta$$

(axis P)

4 unknown, 4 eq.

(b) About to tip over $\Rightarrow$ Box rotates around Q

For B:

$$\text{Torque: } N_1 L \cos \theta \leq G_B \cdot \frac{1}{2} b$$

(axis Q)

For L:

$$\text{Torque: } N_1 L \cos \theta = G_L \cdot \frac{L}{2} \sin \theta + G_M \cdot h \tan \theta$$

(axis P)

$$\Rightarrow h \leq \frac{m_B b - m_L L \sin \theta}{2 m_M \tan \theta}$$

$$\mu_s \geq \frac{m_L L \sin \theta + 2 m_M h \tan \theta}{2 m_B L \cos \theta}$$

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Work in Rotational Motion

Definition and Formula

$$W_{12} = \int_{\theta_1}^{\theta_2} \tau_z d\theta$$

$$P = \tau_z \omega_z \quad P = Fv$$

Work-Kinetic Energy Theorem

$$W = K_2 - K_1,$$

where $K = \frac{1}{2} I \omega^2$

if there's only rotation

What if translation + rotation?

(C is CoM) $K = ?$
 $= \frac{1}{2} m v_C^2 + \frac{1}{2} I_C \omega^2$

initial: v_{C1}, ω_1

final: v_{C2}, ω_2

$$W = \frac{1}{2} m (v_{C2}^2 - v_{C1}^2) + \frac{1}{2} I_C (\omega_2^2 - \omega_1^2)$$

Angular Momentum

Definition and Formula

The rate of change of the angular momentum of a particle equals the torque of the net force acting on it.

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

⇒ See more VM240

Cases for VP150
↑

Especially, when for rotation about an axis of symmetry, or for a planar object contained in the x-y plane:



② Disk (no thickness)
Bar ...



Conservation of Angular Momentum

Definition and Formula

When the net external torque on a system is zero, then the total angular momentum of the system is conserved.

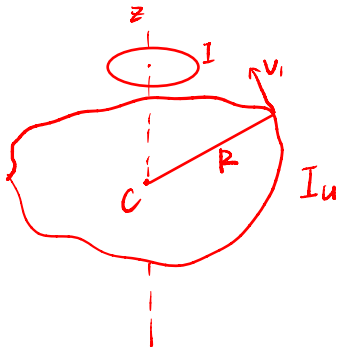
$$\bar{L}_z = I_z \bar{\omega}_z = \text{const} \quad \Rightarrow \quad \vec{p} = m \vec{v}_{cm} \\ = \text{const} \\ \text{if no external F.}$$

Exercise

(5 points) A certain object with unknown shape is rotating about a fixed vertical axis passing through its center of mass. A scientist measures the part of the object farthest from the object's center of mass to be moving with speed v_1 at radius R_1 . To measure the moment of inertia a scientist drops a stationary disk with $I = \frac{1}{2} M_d R_d^2$ on top of the unknown object. The disk is dropped such that its axis of symmetry (perpendicular to the disk's plane) is collinear with the axis about which the unknown object is rotating. The two objects hit and stick together and rotate as one object following the collision. With the addition of the disk the angular speed is found to decrease to 50% of its original value.

If the mass of the unknown object is equal to $\frac{1}{2} M_d$ and $R_d = R_1$, what is its moment of inertia?

no use.



$$\omega_0 = \frac{v_1}{R}$$

$$\text{initial state: } L_i = I_u \cdot \omega_0$$

$$\text{final state: } L_f = (I_u + I) \cdot \frac{1}{2} \omega_0$$

$$\Rightarrow L_i = L_f \Rightarrow I_u = I = \frac{1}{2} M_d R_d^2$$

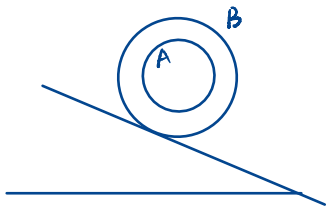
Exercise

(5 points) A uniform hollow cylinder has mass M , inner radius r and outer radius R . Another uniform cylinder with mass m and radius r is placed inside the hollow cylinder. Initially, the filled cylinder is held at rest at the top of a plane inclined at an angle α to the horizontal. It is then let to roll down the plane without slipping.

We repeat the experiment in two versions

- (a) the inner cylinder is permanently glued to the hollow one,
- (b) the inner cylinder is free to rotate without friction inside the hollow cylinder.

In which case is the time needed for the cylinder to roll to the bottom of the incline plane shorter? Justify your answer without calculations.



$$\Delta U_B = \Delta K = \Delta K_t + \Delta K_r$$

Which case has larger I ?

$$\Rightarrow I_0 = I_A + I_B > I_B$$

$$I_B = I_B$$

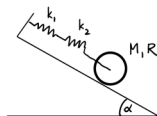
$$\Delta K_r = \frac{1}{2} I \omega^2 \quad K_t = \frac{1}{2} m v^2 \quad \omega R = v$$

$\Rightarrow$ (b)

(a) requires more energy to reach
the same translational speed

Exercise

(15 points) A uniform ball with mass M and radius R is placed on a long plane, inclined at the angle α to the horizontal. A yoke passing through the center of the ball is connected to two springs, obeying Hooke's law with spring constants k_1 and k_2 , connected in series. The other end of the spring is connected to a fixed support at the top of the incline. The ball is released when the springs are neither stretched nor compressed. There is no slipping of the ball on the inclined plane at any time. The acceleration due to gravity g is known.



$$k = \frac{k_1 k_2}{k_1 + k_2}$$

- How far down the plane does the ball travel until it stops instantaneously?
What is the maximum speed of the center of mass of the ball?
- What is the direction of the force of friction between the plane and the ball at any instant of time?

- Derive an equation for the acceleration of the center of mass of the ball in terms of k_1 , k_2 , M , R , g , α , and the distance of the center of mass of the ball x , measured from its initial position. set origin at L_1 for U_0

$$E_{\text{tot}} = \frac{1}{2} M v^2 + \frac{1}{2} \cdot \frac{2}{5} M R^2 \left(\frac{v}{R} \right)^2 + M g (L_1 - x) \sin \alpha + \frac{1}{2} k x^2$$

$$= M g L_1 \sin \alpha \quad (\text{at initial state})$$

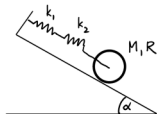
$$\Rightarrow \frac{7}{10} M v^2 = -\frac{1}{2} k \left(x - \frac{1}{k} M g \sin \alpha \right)^2 + \frac{1}{2k} M^2 g^2 \sin^2 \alpha$$

$$\Rightarrow v_{\text{max}} = \sqrt{\frac{5 M (k_1 + k_2)}{7 k_1 k_2}} g \sin \alpha \quad \text{when } x = \frac{1}{2} L_1$$

$$\begin{aligned} (a) \quad & M g h = \frac{1}{2} k L_1^2 \\ & h = L_1 \sin \alpha \\ \Rightarrow & L_1 = 2 M g \frac{k_1 + k_2}{k_1 k_2} \sin \alpha \end{aligned}$$

Exercise

(15 points) A uniform ball with mass M and radius R is placed on a long plane, inclined at the angle α to the horizontal. A yoke passing through the center of the ball is connected to two springs, obeying Hooke's law with spring constants k_1 and k_2 , connected in series. The other end of the spring is connected to a fixed support at the top of the incline. The ball is released when the springs are neither stretched nor compressed. There is no slipping of the ball on the inclined plane at any time. The acceleration due to gravity g is known.



- How far down the plane does the ball travel until it stops instantaneously?
What is the maximum speed of the center of mass of the ball?
- What is the direction of the force of friction between the plane and the ball at any instant of time?
- Derive an equation for the acceleration of the center of mass of the ball in terms of k_1 , k_2 , M , R , g , α , and the distance of the center of mass of the ball x , measured from its initial position.

(b) according to (a):

$$0 < x < \frac{1}{2}L:$$

$$v \uparrow \omega \uparrow$$

$$\Rightarrow \beta > 0 \Rightarrow f \text{ upwards along inclined plane}$$

$$\frac{1}{2}L < x < L:$$

$$v \downarrow \omega \downarrow \Rightarrow \beta < 0 \Rightarrow f \text{ downwards ...}$$

$$(c) \quad 0 < x \leq \frac{1}{2}L$$

$$\begin{cases} Ma = mgsin\alpha - kx - f_s \\ f_s R = I\beta = \frac{2}{5}MR^2 \cdot \frac{a}{R} \end{cases}$$

$$\Rightarrow a = \frac{5gsin\alpha}{7} - \frac{5}{7M} \frac{k_1 k_2}{k_1 + k_2} x$$

$$\frac{1}{2}L < x < L:$$

$$\begin{cases} Ma = \dots + f_s \\ f_s R = -I\beta \end{cases}$$

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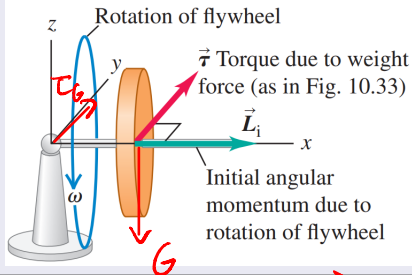
4 Equilibrium

Precession

Working Principle

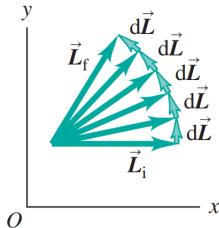
(a) Rotating flywheel

When the flywheel is rotating, the system starts with an angular momentum $\vec{L}_i$ parallel to the flywheel's axis of rotation.



(b) View from above

Now the effect of the torque is to cause the angular momentum to precess around the pivot. The gyroscope circles around its pivot without falling.



$$\vec{\tau}_G = \frac{d\vec{L}}{dt}$$

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Equilibrium

Equilibrium Equations

Equation for torque equilibrium (rotational motion)

Equation for force equilibrium (translational motion)

Reference

Zhemin Qu, 2023SU VP150 RC.

Mateusz Krzyzosiak, 2024SU VP150 Lec.